

Applied Thermo-fluids II

3-1-0

Basic Refrigeration Cycles- Reverse Rankine, Reverse Brayton and Reverse Stirling Gas cycle refrigeration systems Actual vapour compression refrigeration cycle, first and second law analysis, superheating, sub-cooling and liquid suction heat exchanger Refrigerants and their properties Refrigeration system components Vapour Absorption Refrigeration system (Water-LiBr) [16 hours]

IC Engines- Components of an engine, 2-stroke and 4-stroke engine cycle, four stroke SI and CI engine cycles, Thermal and mechanical efficiency, combustion efficiency, volumetric efficiency, compression ratio, torque, power, mean effective pressure, specific fuel consumption, Air/fuel ratio, Cycles revision Valve timing diagram, Ignition system, Air-fuel mixing Combustion in SI engine and CI engine Engine Exhaust Emission Alternative Fuels [16 hours]

Power Plant- Fossil fuel based steam power plant, steam generators, HRSG, fluidized bed boiler, draft and fan, cold water system. Cycles: Reheat and regeneration in Rankine cycle; types of feed heaters, deaerator, Combined cycle , cogeneration, super critical cycle, Losses and practical issues. Steam turbine: Convergent and convergent-divergent steam nozzles, impulse and reaction turbine; compounding, losses in turbine. [16 hours]

Text Books: 1. W.F. Stoecker and J.W. Jones, Refrigeration and Air-conditioning, McGraw Hill, 1982. 2. Internal Combustion Engines, Ganesan, V. 3. Power Plant Engineering, 4th Edition, P. K. Nag, Tata McGraw Hill Power Plant Engineering Theory and Practice. D. K. Mondal, S. Chakraborty, A. K. Das, P. K. Das, Wiley India

Reference Books: 1. C.P. Arora, Refrigeration and Air-conditioning, Tata McGraw Hill, 2009, 2. R.C. Arora, Refrigeration and Air-conditioning, Prentice Hall India, 2010 3. Internal Combustion Engine, Johan B Heywood. 4. Power Plant Technology, M. M. El-Wakil, McGraw Hill Steam its Generation and Use, Babcock and Wilcox